

Khuong Ngo

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EDUCATION

Missouri State University

B.S. Computer Science, GPA: 3.82/4.0

Coursework: Data Structures & Algorithms, Artificial Intelligence, Computer Networks

Dean's List: Spring/Fall 2023, 2024; Spring 2025

Springfield, MO

Aug 2022 – Jun 2026

TECHNICAL SKILLS

- **Languages:** C++, C#, Java, Python
- **Systems & Backend:** Multithreading, WebSocket, TCP/IP, Concurrency
- **Databases:** SQL Server, MySQL, MongoDB (**ACID transactions**)
- **Frameworks & Tools:** .NET WinForms, Node.js, Django, Git

PROJECTS

MeetSummarizer

July 2025 – Nov 2025

Source

- Built a **Speech-to-Text (STT)** pipeline using **Transformers.js** and **WebGPU**, achieving near-instant transcription by offloading **Whisper-small** inference to the client-side GPU.
- Improved transcription accuracy using a sliding-window audio buffer with a **3-second overlap**, preventing word truncation at 15-second chunk boundaries.
- Leveraged **WebRTC** for **low-latency** peer-to-peer video streaming, supporting dynamic device switching and bandwidth-efficient media track management.
- Designed a **privacy-by-design** architecture where raw audio never leaves the user's device, ensuring enterprise-grade data security.

Flappy Bird AI Agent

July 2025 – Dec 2025

Demo

- Developed an autonomous Flappy Bird agent using **genetic algorithms** and **neural networks** in **JavaScript** with **p5.js**, training agents to navigate randomized obstacle patterns.
- Implemented configurable evolution controls for **population size**, **mutation rate**, and **network architecture**, improving learning stability across generations.
- Tuned **fitness scoring** and **reward shaping** to produce robust, collision-free gameplay under varied starting conditions.

Inventory Management System

Jan 2025 – Mar 2025

- Developed a **multithreaded** desktop application for concurrent inventory operations using **C#** and **.NET WinForms**, supporting 5+ simultaneous users.
- Implemented **ACID-compliant SQL Server transactions** and **fine-grained locking** to ensure data integrity under high concurrency.
- Built real-time dashboards and dynamic charting for live inventory metrics; optimized update pipelines to keep **UI latency** under user-perceptible thresholds.
- Added unit/integration tests and stress tests to validate behavior under concurrent workloads and failure scenarios.

LEADERSHIP & INVOLVEMENT

Vietnamese Student Association

Missouri State University

Active Member, Aug 2022 – Present

- Promoted Vietnamese culture through community events and cultural exchange programs; coordinated logistics for campus events.